A Comparative Analysis of Two Capacitor-Less LDO Topologies in 130nm CMOS Technology

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Need for Low Dropout Regulators

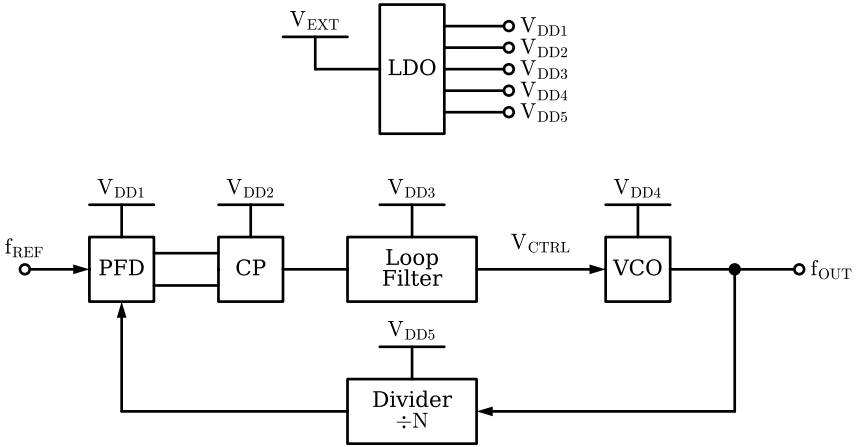


Figure: Phase Locked Loop with LDO

Target Specifications

Parameter	Target Value
Power Supply Voltage (V_{EXT})	3 V
Bias Current (I_{bias})	5 µA
Load Current (I_L)	500 µA
Load Capacitance (C_L)	50 pF
Saturation Margin ($V_{DSAT-MARG}$)	$>$50 mV
Reference Voltage (V_{REF})	1.2 V $\pm$ 1 %
Output Voltages (V_{OUT})	2.3 V, 2.1 V, 1.6 V
Compensation Capacitor (C_c)	$<$ 7 pF
Phase Margin @ No-Load	$>$ 60°
Phase Margin @ Full-Load	$>$ 70°
Loop Gain @ 1 Hz	$>$ 50 dB
PSRR @ 1 Hz	$>$ 65 dB
PSRR @ 1 MHz	$>$ 2 dB
PSRR @ 10 MHz	$>$ 2 dB
Output-Referred Noise @ 10 Hz	$< -115\,\text{dBV}/\sqrt{\text{Hz}}$
Output-Referred Noise @ 1 kHz	$< -125\,\text{dBV}/\sqrt{\text{Hz}}$
Output-Referred Noise @ 10 kHz	$< -134\,\text{dBV}/\sqrt{\text{Hz}}$
Output-Referred Noise @ 100 kHz	$< -143\,\text{dBV}/\sqrt{\text{Hz}}$
Output-Referred Noise @ 1 MHz	$< -155\,\text{dBV}/\sqrt{\text{Hz}}$

Table: Technical LDO Target Specifications

PMOS LDO Schematic

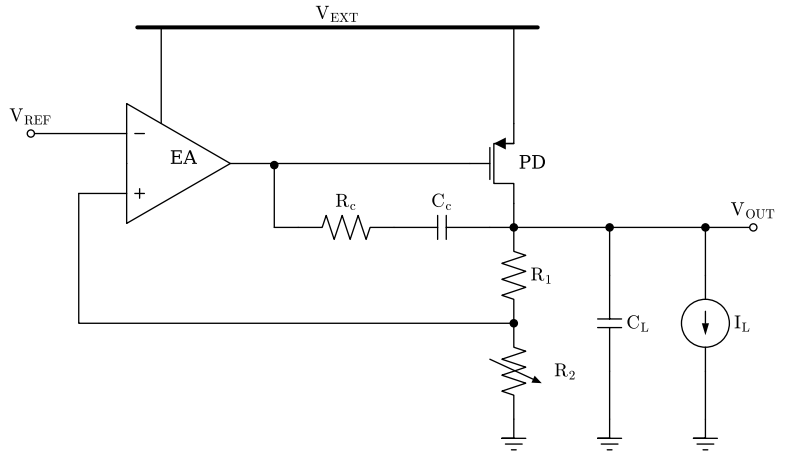


Figure: General LDO schematic

LDO's Small-Signal Model

Output voltage

$$\frac{V_{OUT}}{V_{REF}} \approx \frac{1}{\beta}, \qquad \frac{R_1}{R_2} = \frac{V_{REF}}{V_{OUT}} - 1$$

Stability

$$\omega_{p1} = \frac{1}{g_{m,PD} R_1 R_2 C_c}, \quad \omega_{p2} = \frac{g_{m,PD}}{C_1 + C_2}, \quad \omega_z = \frac{-1}{C_c(1/g_{m,PD} - R_c)}$$

PSRR

$$|PSRR_{DC}| \cong \frac{1}{\beta A_{EA,0} g_{m,PD} r_{ds,p}}$$

Output-referred noise

$$S_{n,o}(f) \cong (S_{n,ref}(f) + S_{n,EA}(f))(1 + \frac{R_1}{R_2})^2 + S_{n,R2}(f)(\frac{R_1}{R_2})^2 + S_{n,R_1}(f)$$

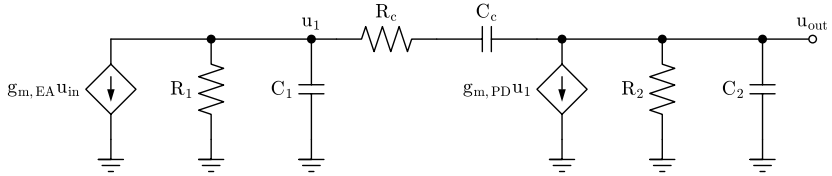


Figure: Small-signal model of the LDO

5T OTA (Simple Differential Pair)

The 5T OTA is a simple differential pair with a gain of:

$$A_{5T} = \frac{g_{m,12}}{g_{5T,out}}$$

while the output impedance is given by:

$$g_{5T,out} = g_{ds,2} + g_{ds,4}$$

As for the output swing, to keep all transistors in saturation, we must have:

$$V_{REF} - V_T \leq V_{OUT} \leq V_{EXT} - V_{DSAT}$$

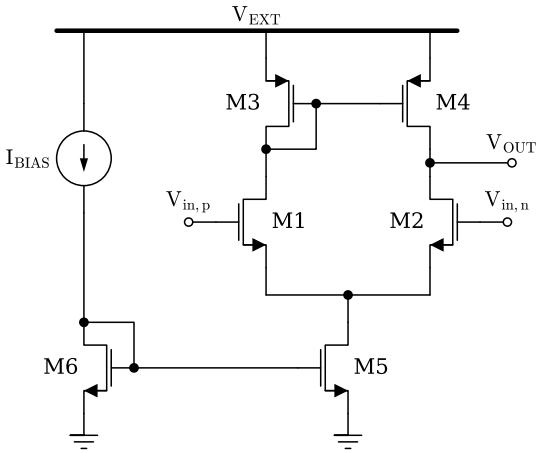


Figure: 5T OTA

5T OTA-Based LDO (Sizing)

Device	Value / Sizing	Unit
M_1, M_2 (Input Pair)	42.0/5.0	µm
M_3, M_4 (Active Load)	3.8/5.0	µm
M_5 (Tail Current)	12.0/5.0	µm
M_6 (Input Current)	5.0/5.0	µm
M_{PD} (Pass Device)	20.0/0.14	µm
R_1 (Divider)	15.00	kΩ
R_{21} (Divider)	16.36	kΩ
R_{22} (Divider)	20.00	kΩ
R_{23} (Divider)	45.00	kΩ
R_c (Compensation)	15.00	kΩ
C_c (Compensation)	7.00	pF

Table: Device Sizing for the 5T OTA LDO

- **Input Pair:** Large W and L to reduce flicker noise and increase loop gain.
- **Active Load:** Large L to increase loop gain; small W to improve PSRR.
- **Tail Current:** Large L for better current mirroring; large W to increase bias current and reduce thermal noise at higher frequencies.
- **Pass Device:** Large W to improve transconductance and loop gain.
- **Compensation:** Nulling resistor and compensation capacitor for improving UGBW and PSRR at higher frequencies.

5T OTA-Based LDO (Simulation Results)

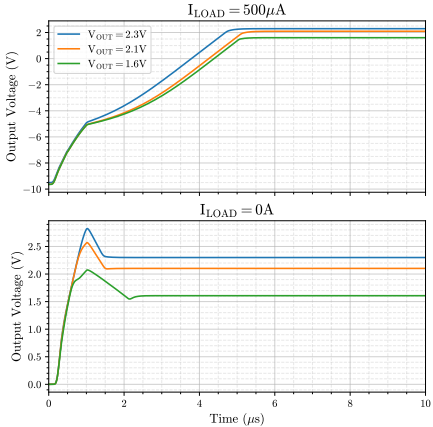


Figure: Ramp Response

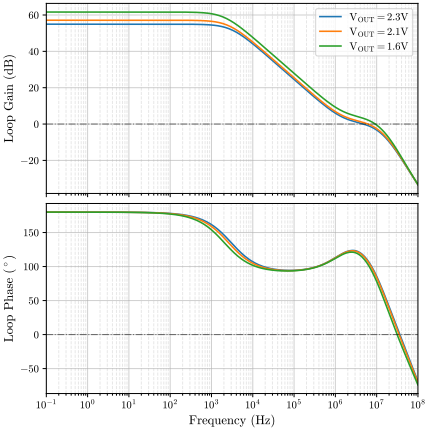


Figure: Bode Plot

5T OTA-Based LDO (Simulation Results)

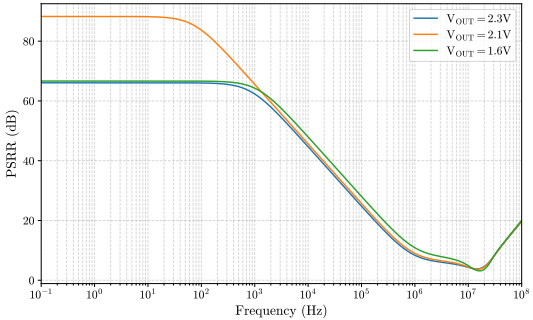


Figure: PSRR

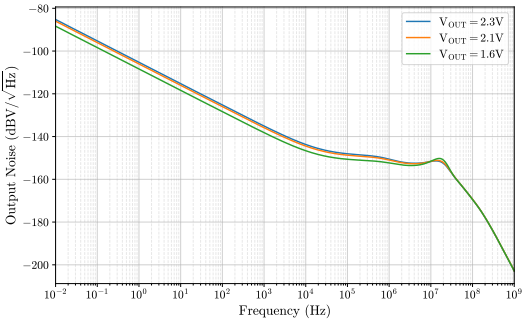


Figure: Output-Referred Noise

Telescopic OTA

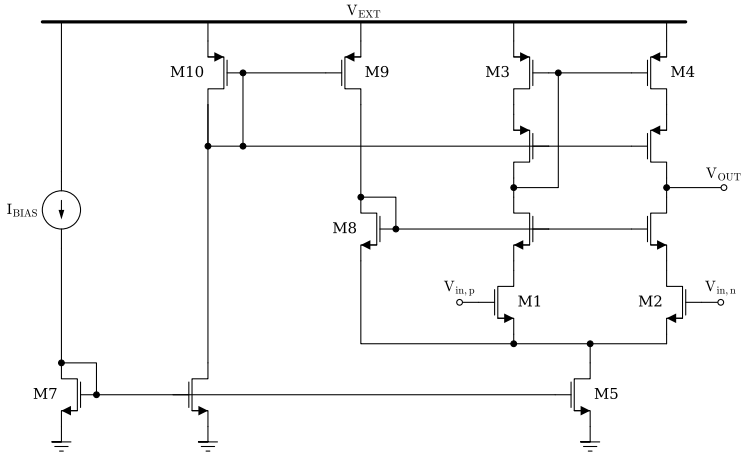


Figure: Telescopic OTA

Telescopic OTA

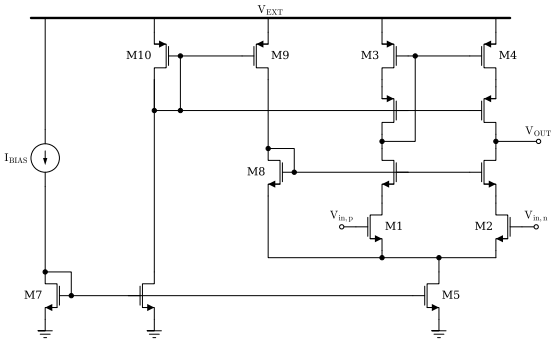


Figure: Telescopic OTA

This leads to the following output swing:

$$V_{REF} - V_T + V_{DSAT} \leq V_{OUT} \leq V_{EXT} - 2V_{DSAT}$$

Both loop gain and output impedance are substantially increased compared to the 5T OTA. More specifically, the loop gain can be given by:

$$A_{TEL} = \frac{g_{m,12}}{g_{tel,out}}$$

whereas the output impedance is given by:

$$g_{tel,out} = g_{ds,2}\frac{g_{ds,2C}}{g_{m,2C}} + g_{ds,4}\frac{g_{ds,4C}}{g_{m,4C}}$$

Bias transistors should be sized as follows to ensure maximum output swing:

$$\frac{W_8}{L_8} = \frac{W_{3,4}}{4L_{3,4}}, \quad \frac{W_{10}}{L_{10}} = \frac{W_{1,2}}{4L_{1,2}}$$

Telescopic OTA-Based LDO (Sizing)

- **Input Pair & Active Load:** Large W and L to reduce flicker noise and increase loop gain.
- **Cascode Transistors:** Small W to ensure saturation; no need for larger transconductance.
- **Current Mirror:** Large L for better current mirroring; small W to keep all devices in saturation while improving PSRR.
- **Pass Device:** Large W to improve transconductance and loop gain.
- **Divider:** Small resistors to minimize noise and decrease the pass device's gate voltage at no-load.
- **Compensation:** Nulling resistor and compensation capacitor for improving UGBW and PSRR at higher frequencies.

Device	Value / Sizing	Unit
M_1, M_2 (Input Pair)	25.0/5.0	µm
M_{1C}, M_{2C} (NMOS Cascode)	0.65/1.0	µm
M_{3C}, M_{4C} (PMOS Cascode)	2.0/1.0	µm
M_3, M_4 (Active Load)	40.0/5.0	µm
M_8 (NMOS Bias)	0.65/4.0	µm
M_9, M_{10} (PMOS Bias)	2.0/4.0	µm
M_5 (Tail Current)	0.95/1.0	µm
M_6 (Bias Current)	0.28/1.0	µm
M_7 (Input Current)	0.28/1.0	µm
M_{PD} (Pass Device)	40.0/0.13	µm
R_1 (Divider)	22.00	kΩ
R_{21} (Divider)	24.00	kΩ
R_{22} (Divider)	29.33	kΩ
R_{23} (Divider)	66.00	kΩ
R_c (Compensation)	20.00	kΩ
C_c (Compensation)	7.00	pF

Table: Device Sizing for the Telescopic OTA LDO

Telescopic OTA-Based LDO (Simulation Results)

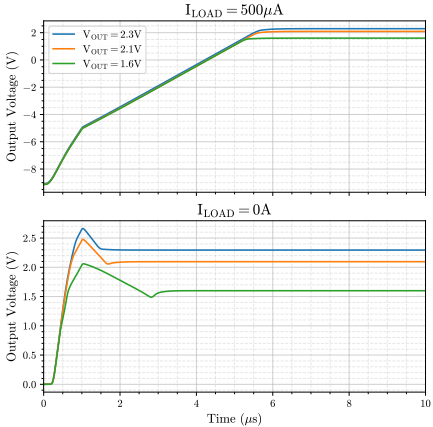


Figure: Ramp Response

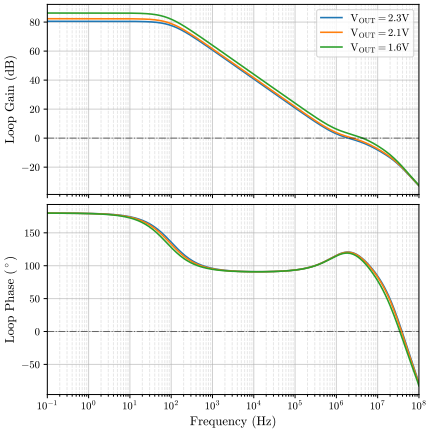


Figure: Bode Plot

Telescopic OTA-Based LDO (Simulation Results)

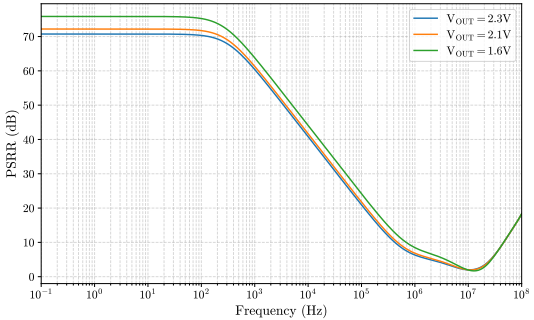


Figure: PSRR

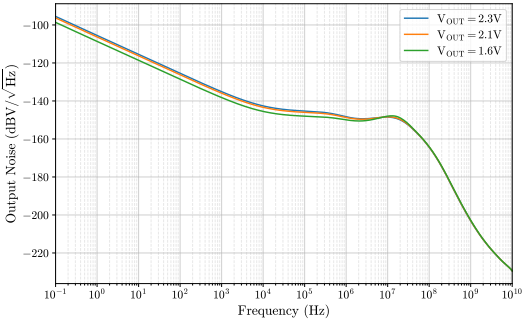


Figure: Output-Referred Noise

Simulation Results of Both Topologies

Parameter	Target Specification	5T OTA (Min – Max)	Telescopic-OTA (Min – Max)
Trim Voltage 1	$= 2.3\,\text{V}$	2.293 – 2.299	2.294
Trim Voltage 2	$= 2.1\,\text{V}$	2.096 – 2.102	2.096
Trim Voltage 3	$= 1.6\,\text{V}$	1.602 – 1.607	1.6
Quiescent Current	$< 100\,\mu\text{A}$	48.16 – 94.49	52.56 – 84.12
Phase Margin ($I_{load} = 0$)	$> 60^\circ$	77.62 – 99.2	74.15 – 99
Phase Margin ($I_{load} = 500\,\mu\text{A}$)	$> 70^\circ$	82.7 – 116.2	107.2 – 120.1
Loop Gain @ 1 Hz	$> 50\,\text{dB}$	54.95 – 61.63	80.53 – 86.23
PSRR @ 1 Hz	$> 65\,\text{dB}$	66.07 – 88.27	70.73 – 75.86
PSRR @ 1 MHz	$> 2\,\text{dB}$	8.43 – 10.95	6.373 – 8.537
PSRR @ 10 MHz	$> 2\,\text{dB}$	4.353 – 4.934	2.107 – 2.118
Output-Referred Noise @ 10 Hz	$< -115\,\text{dBV}/\sqrt{\text{Hz}}$	-118.3 – -115.1	-118.6 – -115.5
Output-Referred Noise @ 1 kHz	$< -125\,\text{dBV}/\sqrt{\text{Hz}}$	-138.1 – -135	-138.1 – -135.1
Output-Referred Noise @ 10 kHz	$< -134\,\text{dBV}/\sqrt{\text{Hz}}$	-146.5 – -143.6	-145.4 – -142.5
Output-Referred Noise @ 100 kHz	$< -143\,\text{dBV}/\sqrt{\text{Hz}}$	-150.6 – -148	-148 – -145.3
Output-Referred Noise @ 1 MHz	$< -155\,\text{dBV}/\sqrt{\text{Hz}}$	-152.3 – -150.5	-149.9 – -148.3

Table: Simulated Performance Summary (Across All Trims)

Comparison

- **5T OTA-Based LDO:** Offers the most balanced results. Should be preferred when:
  1. A smaller pass device is needed.
  2. Better error amplifier output swing is required to keep all devices in saturation.
  3. A simpler design with fewer transistors is desired.
  4. Slightly better noise performance at high frequencies is needed.
  5. Slightly better PSRR performance at high frequencies is needed.
- **Telescopic OTA-Based LDO:** Maximizes loop gain and PSRR. Should be preferred when:
  1. A larger pass device can be utilized.
  2. The limited output swing of the error amplifier is not a concern.
  3. More complex designs with additional transistors are acceptable.
  4. Higher PSRR values at low frequencies are required.
  5. Better phase margin under load is needed.

Conclusion and Future Work

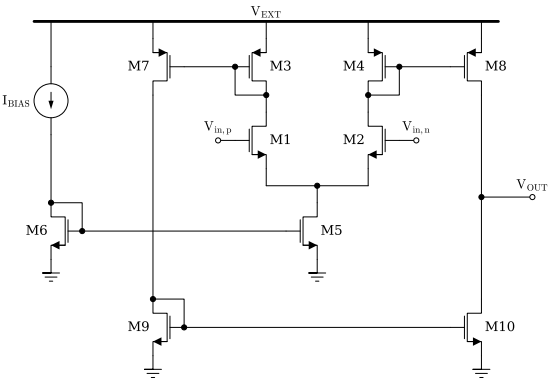


Figure: Current Mirror OTA

Both designs perform well across almost all specified performance targets, with the exception of the output-referred noise specification at higher frequencies. Meeting the noise specification at higher frequencies would require increasing the input-transistor bias current, as thermal noise dominates in this regime. However, increasing the bias current in both topologies reduces the output impedance, thereby decreasing loop gain and PSRR performance. Alternative architectures with separate input and gain stages, each optimized for different operating currents, warrant further investigation.

Current Mirror OTA

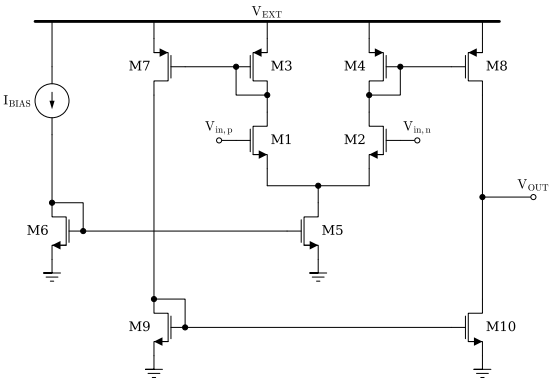


Figure: Current Mirror OTA

Loop gain can be improved by mirroring factor B:

$$A_{CM} = \frac{B g_{m,12}}{g_{cm,out}}$$

whereas the output impedance is given by:

$$g_{cm,out} = g_{ds,8} + g_{ds,10}$$

Current Mirror OTA-Based LDO (Sizing)

Device	Value / Sizing	Unit
M_1, M_2 (Input Pair)	40.0/0.5	µm
M_3, M_4 (Active Load)	5.0/5.0	µm
M_7, M_8 (PMOS CM)	2.5/5.0	µm
M_9, M_{10} (NMOS CM)	1.0/5.0	µm
M_5 (Tail Current)	16.0/1.0	µm
M_6 (Bias Current)	8.0/1.0	µm
M_{PD} (Pass Device)	40.0/0.5	µm
R_1 (Divider)	22.00	kΩ
R_{21} (Divider)	24.00	kΩ
R_{22} (Divider)	29.33	kΩ
R_{23} (Divider)	66.00	kΩ
R_c (Compensation)	15.00	kΩ
C_c (Compensation)	7.00	pF

Table: Device Sizing for the Current Mirror OTA-Based LDO

Current Mirror OTA-Based LDO (Simulation Results)

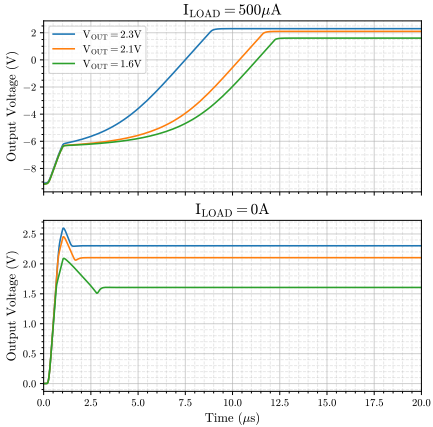


Figure: Ramp Response

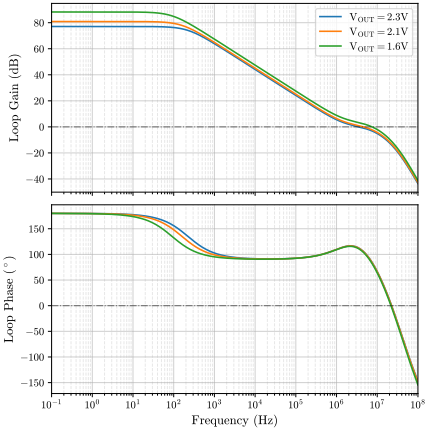


Figure: Bode Plot

Current Mirror OTA-Based LDO (Simulation Results)

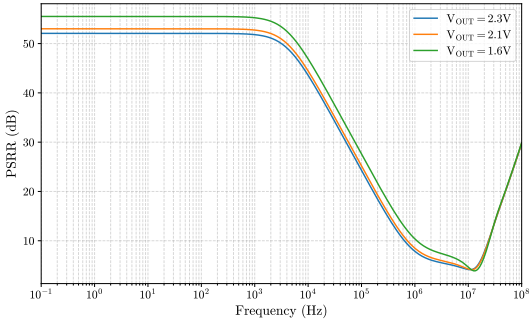


Figure: PSRR

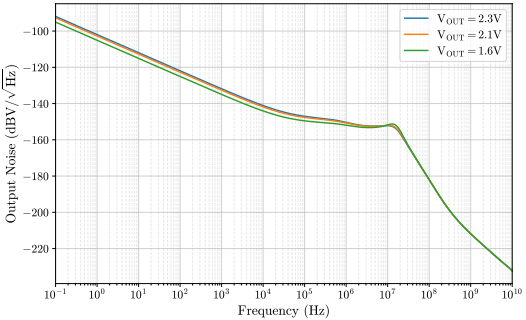


Figure: Output-Referred Noise

Current Mirror OTA-Based LDO Simulation Results

Parameter	Target Specification	Current Mirror OTA (Min – Max)
Trim Voltage 1	$= 2.3\,\text{V}$	2.301 – 2.303
Trim Voltage 2	$= 2.1\,\text{V}$	2.103 – 2.104
Trim Voltage 3	$= 1.6\,\text{V}$	1.605 – 1.606
Quiescent Current	$< 100\,\mu\text{A}$	54.26 – 85.94
Phase Margin ($I_{load} = 0$)	$> 60^\circ$	64.39 – 85.28
Phase Margin ($I_{load} = 500\,\mu\text{A}$)	$> 70^\circ$	82.14 – 112.3
Loop Gain @ 1 Hz	$> 50\,\text{dB}$	77.09 – 88.23
PSRR @ 1 Hz	$> 65\,\text{dB}$	52.07 – 55.5
PSRR @ 1 MHz	$> 2\,\text{dB}$	7.923 – 10.45
PSRR @ 10 MHz	$> 2\,\text{dB}$	4.337 – 4.802
Output-Referred Noise @ 10 Hz	$< -115\,\text{dBV}/\sqrt{\text{Hz}}$	-115 – -111.9
Output-Referred Noise @ 1 kHz	$< -125\,\text{dBV}/\sqrt{\text{Hz}}$	-134.9 – -131.8
Output-Referred Noise @ 10 kHz	$< -134\,\text{dBV}/\sqrt{\text{Hz}}$	-144 – -141
Output-Referred Noise @ 100 kHz	$< -143\,\text{dBV}/\sqrt{\text{Hz}}$	-149.5 – -147
Output-Referred Noise @ 1 MHz	$< -155\,\text{dBV}/\sqrt{\text{Hz}}$	-151.9 – -150.3

Table: Simulated Performance Summary (Across All Trims)

Acknowledgements

Thank You!

Any Questions?

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